**LAB-ASSIGNMENT-1**

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**ENROLLMENT NO:S25MCAG0010**

1. Create a database named as Redbus online ticket booking system.

```
Command Prompt - My
Microsoft Windows [Version 10.0.26200.7623]
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C:\Users\Ankit kumar>Mysql -u root -p
Enter password: *********
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 18
Server version: 8.0.44 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
mysql> CREATE DATABASE redbus;
ERROR 1007 (HY000): Can't create database 'redbus'; database exists
mysql> CREATE DATABASE redbus;
Query OK, 1 row affected (0.02 sec)

mysql> |
```

2. Use Redbus to enter in created Database

```
Command Prompt - My

Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

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Enter password: *********
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mysql>
mysql> CREATE DATABASE redbus;
ERROR 1007 (HY000): Can't create database 'redbus'; database exists
mysql> CREATE DATABASE redbus;
Query OK, 1 row affected (0.02 sec)

mysql> USE redbus;
Database changed
mysql>
```

3. Create three tables named as List_buses, bus_status and passengers.

```
Command Prompt - My

Database changed
mysql> CREATE TABLE list_buses (
    ->  bus_no VARCHAR(10),
    ->     bus_name VARCHAR(50),
    ->     source VARCHAR(50),
    ->     destination VARCHAR(50),
    ->     fare INT,
    ->     date DATE,
    ->     time TIME
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE bus_status (
    ->     bus_no VARCHAR(10),
    ->     available_seats INT,
    ->     booked_seats INT
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE passengers (
    ->     passenger_id VARCHAR(10),
    ->     p_name VARCHAR(50),
    ->     bus_no VARCHAR(10),
    ->     gender VARCHAR(2),
    ->     age INT,
    ->     status VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

4. Display all columns and data of the table passengers.

```
mysql> SELECT * FROM Passengers;
+--------------+----------+--------+--------+------+---------+
| passenger_id | p_name   | bus_no | gender | age  | status  |
+--------------+----------+--------+--------+------+---------+
| SCS012       | Arjun    | UP301  | M      |   17 | Confirm |
| SCS013       | Anamika  | HR302  | F      |   15 | Confirm |
| SCS018       | Divya    | UP101  | F      |   65 | Waiting |
| SCS014       | Diya     | HR303  | F      |   19 | Confirm |
| SCS015       | Abhishek | UP505  | M      |   57 | Confirm |
| SCS090       | Shiva    | DL701  | M      |   16 | Waiting |
| SCS071       | Rahul    | DL306  | M      |   21 | Confirm |
| SCS043       | Rupam    | UP501  | F      |   22 | Confirm |
| SCS017       | Hina     | UP301  | F      |   23 | Waiting |
| SCS022       | Alam     | HR302  | M      |   21 | Waiting |
| SCS056       | Satya    | UP101  | M      |   23 | Confirm |
+--------------+----------+--------+--------+------+---------+
11 rows in set (0.00 sec)
```

5. Display all records from the passengers table for male passengers.

```
mysql> SELECT *
    -> FROM passengers
    -> WHERE gender = 'M';
+--------------+----------+--------+--------+------+---------+
| passenger_id | p_name   | bus_no | gender | age  | status  |
+--------------+----------+--------+--------+------+---------+
| SCS012       | Arjun    | UP301  | M      |   17 | Confirm |
| SCS015       | Abhishek | UP505  | M      |   57 | Confirm |
| SCS090       | Shiva    | DL701  | M      |   16 | Waiting |
| SCS071       | Rahul    | DL306  | M      |   21 | Confirm |
| SCS022       | Alam     | HR302  | M      |   21 | Waiting |
| SCS056       | Satya    | UP101  | M      |   23 | Confirm |
+--------------+----------+--------+--------+------+---------+
6 rows in set (0.00 sec)

mysql> |
```

6. Find all records in the passengers table with null values in P_name column.

```
mysql> SELECT *
    -> FROM passengers
    -> WHERE p_name IS NULL;
Empty set (0.00 sec)

mysql> |
```

7. Find all the female passengers whose age is above 18.

```
mysql> SELECT *
    -> FROM passengers
    -> WHERE gender = 'F' AND age > 18;
+--------------+--------+--------+--------+------+---------+
| passenger_id | p_name | bus_no | gender | age  | status  |
+--------------+--------+--------+--------+------+---------+
| SCS018       | Divya  | UP101  | F      |   65 | Waiting |
| SCS014       | Diya   | HR303  | F      |   19 | Confirm |
| SCS043       | Rupam  | UP501  | F      |   22 | Confirm |
| SCS017       | Hina   | UP301  | F      |   23 | Waiting |
+--------------+--------+--------+--------+------+---------+
4 rows in set (0.00 sec)

mysql>
```

8. Find all the buses whose source is either ISBT or Pari Chauk.

```
mysql> SELECT *
    -> FROM list_buses
    -> WHERE source IN ('ISBT', 'Pari Chauk');
+--------+-------------+------------+-------------+------+------------+----------+
| bus_no | bus_name    | source     | destination | fare | date       | time     |
+--------+-------------+------------+-------------+------+------------+----------+
| UP301  | Zing bus    | Pari Chauk | Prayagraj   | 1010 | 2022-08-21 | 21:30:00 |
| UP101  | Blue world  | Pari Chauk | Lucknow     | 1240 | 2022-09-24 | 08:05:00 |
| HR303  | Maheshwaram | ISBT       | Ambala      | 1010 | 2022-09-25 | 07:22:00 |
| UP505  | Goluxury    | ISBT       | Kanpur      | 2250 | 2022-09-27 | 19:38:00 |
| DL701  | Vaishnavi   | ISBT       | Chandigarh  | 1550 | 2022-09-28 | 23:55:00 |
| DL306  | Shatabdi    | ISBT       | Dehradun    | 1007 | 2022-09-29 | 20:45:00 |
| UP501  | Safar       | Pari Chauk | Varanasi    | 1080 | 2022-09-30 | 08:35:00 |
+--------+-------------+------------+-------------+------+------------+----------+
7 rows in set (0.00 sec)

mysql>
```

9. Add a column name as blood group in passengers table.

```
mysql> ALTER TABLE passengers
    -> ADD blood_group VARCHAR(5);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

10. Delete the recently added column in the passengers table.

```
mysql> ALTER TABLE passengers
    -> DROP COLUMN blood_group;
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
```

11. Create a new table new_passengers by coping all data from passengers table. Display all from new_passengers table.

```
mysql> CREATE TABLE new_passengers AS
    -> SELECT * FROM passengers;
Query OK, 11 rows affected (0.16 sec)
Records: 11  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM new_passengers;
+--------------+----------+--------+--------+------+---------+
| passenger_id | p_name   | bus_no | gender | age  | status  |
+--------------+----------+--------+--------+------+---------+
| SCS012       | Arjun    | UP301  | M      |   17 | Confirm |
| SCS013       | Anamika  | HR302  | F      |   15 | Confirm |
| SCS018       | Divya    | UP101  | F      |   65 | Waiting |
| SCS014       | Diya     | HR303  | F      |   19 | Confirm |
| SCS015       | Abhishek | UP505  | M      |   57 | Confirm |
| SCS090       | Shiva    | DL701  | M      |   16 | Waiting |
| SCS071       | Rahul    | DL306  | M      |   21 | Confirm |
| SCS043       | Rupam    | UP501  | F      |   22 | Confirm |
| SCS017       | Hina     | UP301  | F      |   23 | Waiting |
| SCS022       | Alam     | HR302  | M      |   21 | Waiting |
| SCS056       | Satya    | UP101  | M      |   23 | Confirm |
+--------------+----------+--------+--------+------+---------+
11 rows in set (0.01 sec)
```

12. Change datatype of any column of new_passengers table.

```
mysql> ALTER TABLE new_passengers
    -> MODIFY age SMALLINT;
Query OK, 11 rows affected (0.11 sec)
Records: 11  Duplicates: 0  Warnings: 0

mysql> |
```

13. Change any column name of new_passengers table.

```
mysql> ALTER TABLE new_passengers
    -> CHANGE p_name passenger_name VARCHAR(50);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> |
```

14. Delete the passengers from new_passengers table whose age is >50.

```
mysql> DELETE FROM new_passengers
    -> WHERE age > 50;
Query OK, 2 rows affected (0.01 sec)

mysql> |
```

15. Change the age of shiva to 17.

```
mysql> UPDATE passengers
    -> SET age = 17
    -> WHERE p_name = 'Shiva';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```